

# The $F_{ST}$ –LD dissociation: the signature that separates empirical ancestry sorting from a neutral null

Module E working note — July 29, 2026

## Abstract

Across 20 replicate *F. polycтена*  $\times$  *F. aquilonia* hybrid populations, the empirical data and a recombination-matched, haplodiploid *neutral null* (real phased founders, no selection, calibrated so the neutral LD background matches) differ in a way no neutral demography can produce: at ancestry-informative (high-DiagnosticIndex) loci the empirical data show **high differentiation but low linkage disequilibrium**, while the null shows exactly the reverse — **low differentiation but high LD**. This single dissociation, visible both in aggregate and SNP-by-SNP (Fig. 1), is the clean signature of divergent, locus-specific selection on ancestry.

## The observation

We measure two things per SNP and bin them by DiagnosticIndex (DI, how strongly a marker distinguishes the parental species): among-population  $F_{ST}$ , and LD between a SNP and its *directly adjacent* SNP(s) (mean  $r^2$ ; i.e.  $ld_w$  at a one-SNP window). The null is the 20 independent simulated demes, each downsampled to the matching empirical population size.

## Why this is diagnostic of selection, not demography

- **The two observations are one phenomenon.** At a diagnostic locus, populations sort ancestry *divergently and locus-specifically* — different populations fix different loci. That builds high per-locus  $F_{ST}$  and, because the *set* of differentiated populations shuffles from one SNP to the next, the among-population covariance between neighbouring SNPs stays near zero: high  $F_{ST}$  inflates the  $r^2$  denominator while the numerator does not, so the pooled LD collapses (within-population LD  $\approx 0.23$  falls to  $\approx 0.03$  when pooled).
- **The null cannot do both at once.** With no selection, ancestry stays admixed and genome-wide-correlated, giving strong *admixture* LD at diagnostic loci (panel B rises) — but drift cannot differentiate ancestry-informative from uninformative markers, so  $F_{ST}$  is flat across DI (panel A). Every neutral model (one panmictic pool, 20 independent demes, or a founding tree) is DI-blind and gives this flat- $F_{ST}$ /high-LD profile.
- **Incomplete, net-unidirectional, locus-autonomous sorting.** Robustly, most diagnostic loci are only partly sorted ( $\sim 7$  of 20 populations fixed) and unidirectional per locus, with a net aquilonia bias (Module A); the divergence lives in *which* populations have sorted *which* loci — an incidence that is near-independent across loci (effective dimensionality  $\approx 15$  of 20; no shared sorting axis). This locus-autonomy is what a few large selected blocks could not produce, and it is what cancels the LD.
- **Orthogonal to the demographic origin.** Whether the populations are one connected pool that spread (neutral  $F_{ST} \approx 0$ , our reading) or many independent hybridizations, the diagnostic-locus pattern is the same — the neutral loci decide the origin, the diagnostic loci decide the selection.

### Empirical vs neutral null: high $F_{ST}$ but low LD at ancestry-informative loci

Aggregate over all data (A,B) and per-SNP distributions by DI class (C–E). LD throughout = mean  $r^2$  between a SNP and its directly adjacent SNP(s).

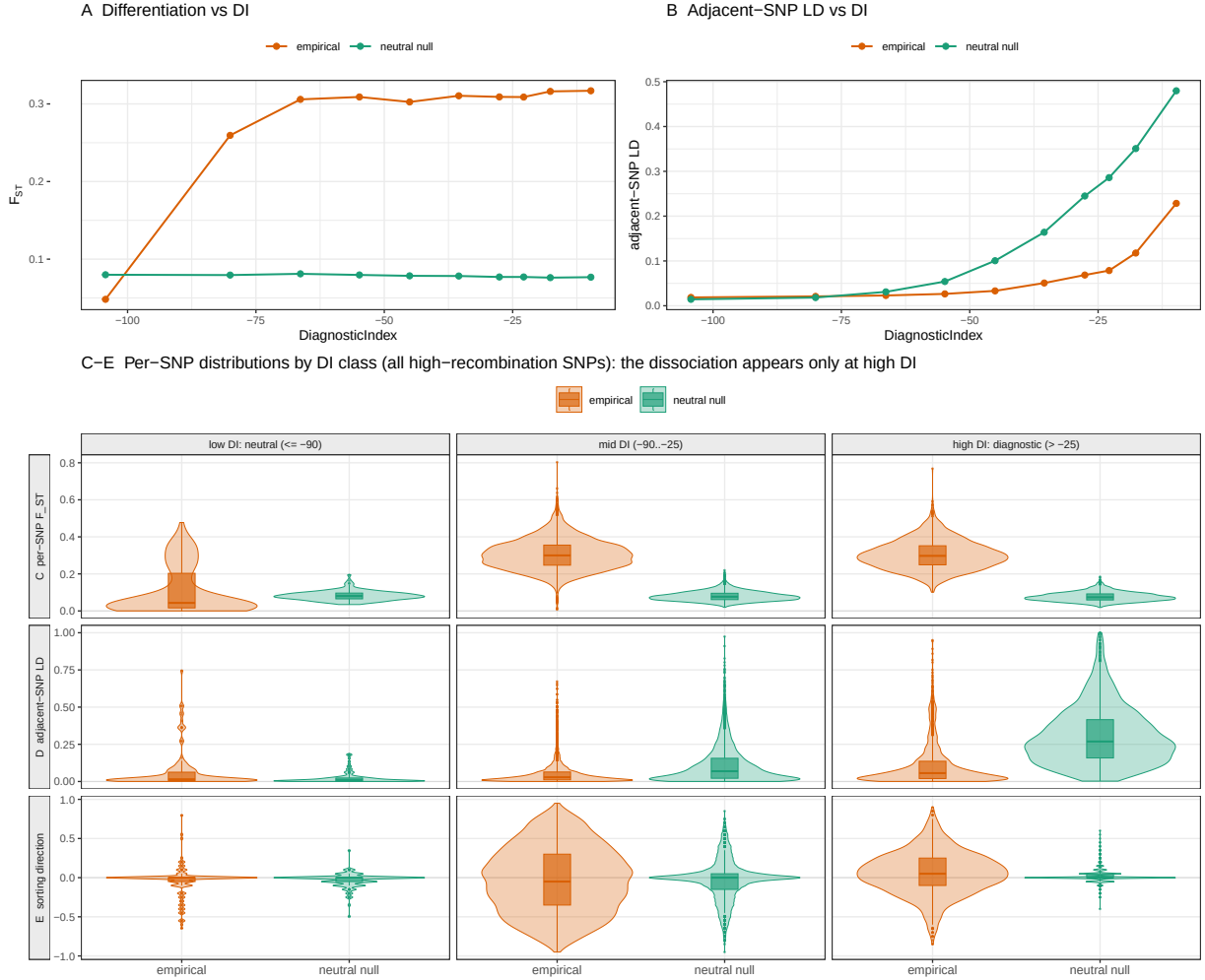


Figure 1: **Empirical vs neutral null.** (A) Among-population  $F_{ST}$  rises with DI in the data but is flat in the null (drift is DI-blind); the null even sits slightly *above* the empirical at neutral loci (it mildly over-differentiates the background), then the empirical pulls far ahead at diagnostic loci. (B) Adjacent-SNP LD does the opposite: it *rises* with DI in the null (admixture LD) while the empirical stays low. (C–E) Per-SNP distributions (violin + quartile box) split by DI class (columns: neutral  $\leq -90$ , mid, diagnostic  $> -25$ ) for the three per-SNP metrics (rows), over all high-recombination SNPs. At *neutral* loci the empirical and null distributions coincide on all three — the calibrated control. The dissociation opens only at mid/high DI: empirical  $F_{ST}$  jumps to  $\sim 0.30$  while the null stays  $\sim 0.08$  (C); adjacent-SNP LD stays low in the empirical but rises in the null, strongest at diagnostic loci (median 0.06 vs 0.27; D); and empirical sorting spreads widely in *mixed directions* while the null sits at zero (E).

## Conclusion

The null is calibrated on the loci where incompatibilities/selection should be absent (neutral background matches), so the diagnostic-locus behaviour is read against a fair baseline. There, the data are simultaneously *more differentiated* and *less linked* than the null — a combination drift, sampling and panmixia are all forbidden from producing. The empirical hybrid populations are therefore undergoing **divergent, locus-specific, largely intrinsic (DI-governed) sorting of parental ancestry**, incompletely and heterogeneously across the range. The  $F_{ST}$ –LD dissociation of Fig. 1 is the most compact evidence for it we have.

Reproduced by `dev/R/moduleE_emp_vs_sim_panels.R`; LD is mean  $r^2$  to the directly adjacent SNP(s) throughout; null = 20 independent demes (450/195,  $K=6250$ , gen 60) downsampled to empirical population sizes.